Data Analysis

Success Hacks

- Be prepared for the class by watching the pre class videos
- Attend every class; Sunday and Thursday
- Participate in class
- Attempt all the MCQs
- Actively advocate for yourself
- Keep your goal in mind
- Consistency is the key

Be patient with yourself !

IK Support Features

- Uplevel - contains all the learning resources - videos, MCQs, reading materials
- Optional Post class videos for additional reference
- Technical coaching sessions on Wednesday to clarify all your queries

INTERVIEW KICKSTART

IK Support Features

- Discord community for peer learning
- TAs to help answer the queries in the live class
- Use academic support (SSAs) for any support/advise you want
- Support tickets - Raise them any time you want any clarifications/ tech concerns , etc

Wednesday and Thursday Sessions

After the main class watch the post class videos – These are purely OPTIONAL

Wednesday - Technical Coaching Session

One hour of topical review on the class. This is an excellent time to reinforce Sunday's content or ask advanced questions, depending on your experience.

Thursday - Assignment Review Session

Two hours to review the MCQs. MCQs are designed to reinforce the learning objectives which tie directly to knowledge and skills you will need in industry and in interviewing.

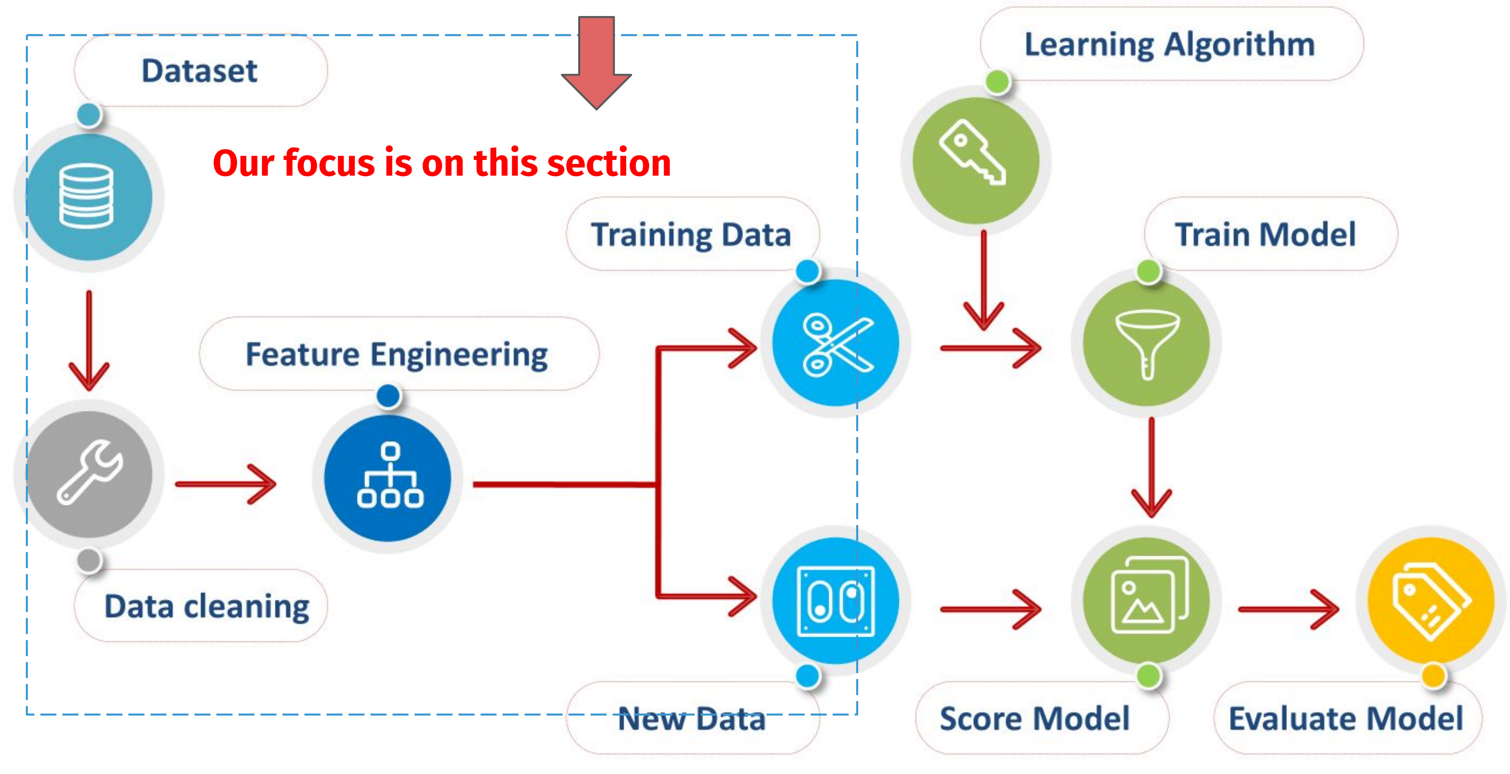
Watched the pre-class foundation videos for this session on Data Analysis?

Watch them before the class for better learning outcomes!

Note: The pre-class materials might be videos or reading material or a combination of both. This might vary as per the content of the module.

The live class materials also might vary as per the module - slides, coding demo or a combination of both.

Machine Learning Pipeline



What is a ‘dataset’?

- A dataset is a collection of values, usually either numbers (quantitative) or strings (qualitative).
- Datasets can include various types of data, such as numbers, text, images, or audio recordings, all stored together for easy access and analysis.
- Datasets are essential in data analytics & ML to identify patterns, draw insights, and train ML models.

Columns

Rows

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Will the dataset look clean always?

No! Not necessarily!

color	director_name	duration	gross	movie_title	anguage	country	budget	title_year	imdb_score
Color	Martin Scorsese	240	116866727	The Wolf of Wall StreetÂ	English	USA	100000000	2013	8.2
Color	Shane Black	195	408992272	Iron Man 3Â	English	USA	200000000	2013	7.2
color	Quentin Tarantino	187	54116191	The Hateful EightÂ	English	USA	44000000	2015	7.9
Color	Kenneth Lonergan	186	46495	MargaretÂ	English	usa	14000000	2011	6.5
Color	Peter Jackson	186	258355354	The Hobbit: The Desolation of SmaugÂ	English	USA	225000000	2013	7.9
	N/A	183	330249062	Batman v Superman: Dawn of JusticeÂ	English	USA	250000000	202	6.9
Color	Peter Jackson	-50	303001229	The Hobbit: An Unexpected JourneyÂ	English	USA	180000000	2012	7.9
Color	Edward Hall	180		RestlessÂ	English	UK		2012	7.2
Color	Joss Whedon	173	623279547	The AvengersÂ	English	USA	220000000	2012	8.1
Color	Joss Whedon	173	623279547	The AvengersÂ	English	USA	220000000	2012	8.1
	Tom Tykwer	172	27098580	Cloud AtlasA	English	Germany	102000000	2012	-7.5
Color	Null	158	102515793	The Girl with the Dragon TattooÂ	English	USA	90000000	2011	7.8
Color	Christopher Spencer	170	59696176	Son of GodÂ	English	USA	22000000	2014	5.6
Color	Peter Jackson	164	255108370	The Hobbit: The Battle of the Five ArmiesÂ	English	New Zealand	250000000	2014	7.5
Color	Tom Hooper	158	148775460	Les MisÃ©rablesÂ	English	USA	61000000	2012	7.6
Color	Tom Hooper	158	148775460	Les MisÃ©rablesÂ	English	USA	61000000	2012	7.6

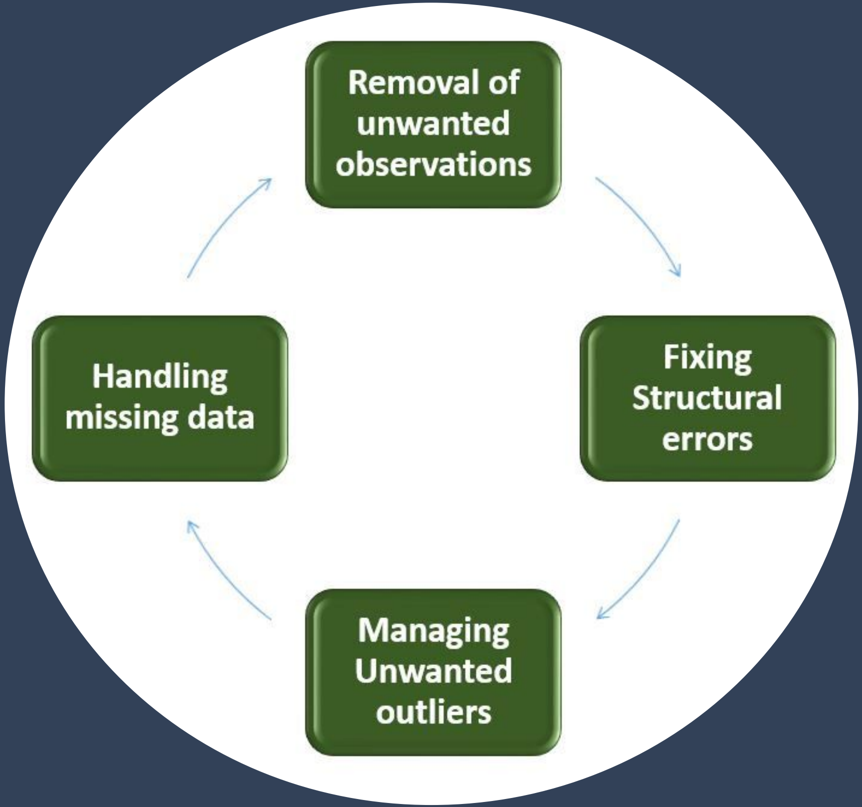
Shall we use the dataset as such without cleaning?

No!

Raw datasets often contain errors, missing values, duplicates, or inconsistencies.

Using uncleaned data can lead to inaccurate results and unreliable models.

How to clean the data?



Missing Values Detection

Missing values

PassengerId	Survived	Pclass	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	male	22	1	0	A/5 21171	7.25		S
2	1	1	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	female	26	0	0	STON/O2. 3101282	7.925		S
4	1	1	female	35	1	0	113803	53.1	C123	S
5	0	3	male	35	0	0	373450	8.05		S
6	0	3	male		0	0	330877	8.4583		Q

Handling missing values

Mobile ID	Mobile Package	Download Speed	Data Limit Usage
1	Fast+	157	80%
2	Lite	99	70%
3	Fast+	167	10%
4	Fast+	N/A	80%
5	Lite	76	70%
6	Fast+	155	10%
7	N/A	N/A	95%
8	Lite	76	77%
9	Fast+	180	N/A

Delete

Delete

Delete

Mobile ID	Mobile Package	Download Speed	Data Limit Usage
1	Fast+	157	80%
2	Lite	99	70%
3	Fast+	167	10%
5	Lite	76	70%
6	Fast+	155	10%
8	Lite	76	77%

Drop Rows with Minimal

Missing Data

Drop Columns with

Many Missing Values

Delete

Mobile ID	Mobile Package	Download Speed	Data Limit Usage
1	Fast+	N/A	80%
2	Lite	N/A	70%
3	Fast+	167	10%
4	Fast+	N/A	80%
5	Lite	76	70%
6	Fast+	N/A	10%
7	Fast+	N/A	95%
8	Lite	76	77%
9	Fast+	180	77%

Mobile ID	Mobile Package	Data Limit Usage
1	Fast+	80%
2	Lite	70%
3	Fast+	10%
4	Fast+	80%
5	Lite	70%
6	Fast+	10%
7	Fast+	95%
8	Lite	77%
9	Fast+	77%

Mean (Download Speed) = 130

Mobile ID	Mobile Package	Download Speed	Data Limit Usage
1	Fast+	157	80%
2	Lite	99	70%
3	Fast+	167	10%
4	Fast+	N/A	80%
5	Lite	76	70%
6	Fast+	155	10%
7	Fast+	N/A	95%
8	Lite	76	77%
9	Fast+	180	95%

Mobile ID	Mobile Package	Download Speed	Data Limit Usage
1	Fast+	157	80%
2	Lite	99	70%
3	Fast+	167	10%
4	Fast+	130	80%
5	Lite	76	70%
6	Fast+	155	10%
7	Fast+	130	95%
8	Lite	76	77%
9	Fast+	180	95%

Impute Missing Values:

Numerical Data - using mean/ median

Categorical Data - using mode

Handling Unwanted values/ Duplicates

Duplicate entries can distort statistical analyses and ML models.

ID	Name	Age	City
1	John	25	New York
2	Alice	30	Los Angeles
3	John	25	New York
4	Bob	22	Chicago
5	Alice	30	Los Angeles

ID	Name	Age	City
1	John	25	New York
2	Alice	30	Los Angeles
4	Bob	22	Chicago

Data Analysis & Visualization

Steps involved in Data Analysis

Univariate Analysis

Focus: One Variable at a time
Goal: To understand the distribution, central tendency (like mean or median), and spread (like range or standard deviation) of that single variable

- Summary Statistics
- Numerical Variables
  - Histogram
  - Violin PLot
  - Box Plot
- Categorical Variables
  - Count Plot
  - Bar Graph
  - Pie Chart

Bivariate Analysis

Focus: Two Variables at a time
Goal: Examining the relationship between two variables to see how they influence each other.

- Numerical Variables
  - Scatter Plot
  - Joint Plot
- Categorical Variables
  - Count Plot
- Numerical vs Categorical
  - Bar Plot
  - Box Plot
  - Violin Plot
  - Displot

Multivariate Analysis

Focus: More than Two Variables at a time
Goal: To understand the complex relationships between the variables

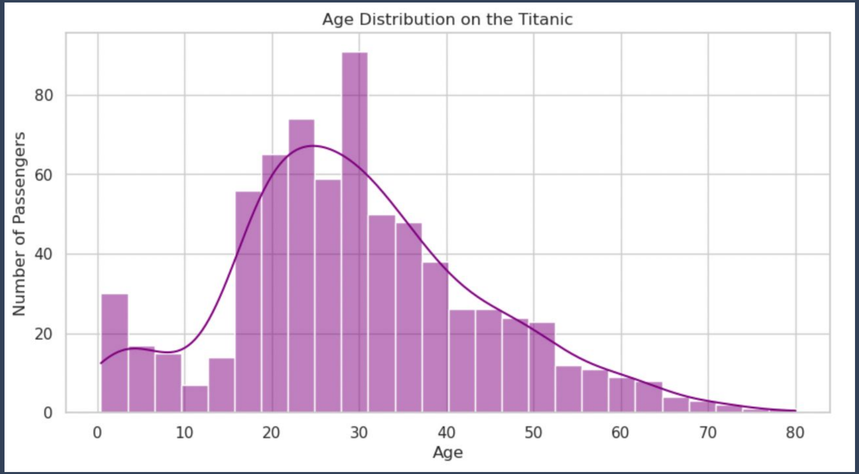
- Bar Plot
- Histogram
- Heat Map
- Pair Plot

Let’s take this example...

Let’s consider the Titanic dataset which contains details about passengers such as age, gender, and ticket class. Using this, we analyze and find out factors that influenced survival during the tragic sinking.

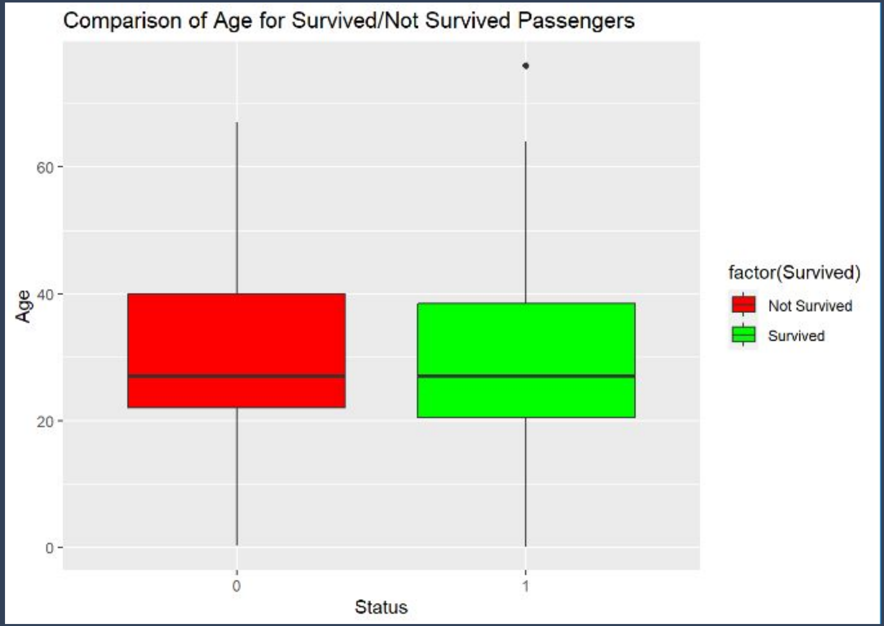
Univariate Analysis for Titanic Data

UNIVARIATE ANALYSIS
What was the age distribution of passengers on the Titanic? - Examines how old the passengers were, from the youngest to the oldest.
How many passengers were in each ticket class? - Looks at the number of passengers who bought first-class, second-class, or third-class tickets.



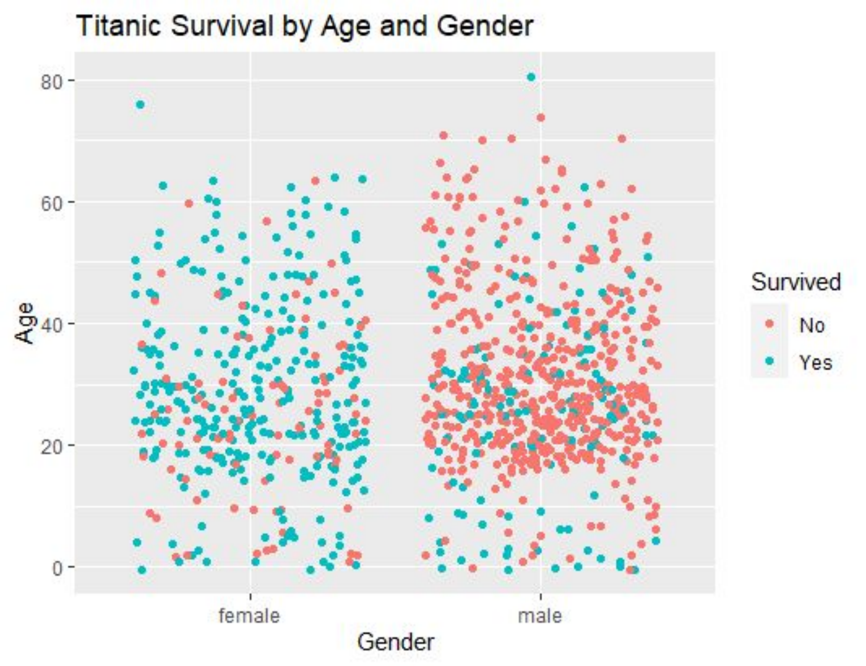
Bivariate Analysis for Titanic Data

BIVARIATE ANALYSIS
Did ticket class influence the chances of survival? - Compares survival rates across different ticket classes to see if those in higher classes had better chances of survival.
Was there a relationship between age and survival? - Investigates whether younger or older passengers were more likely to survive.



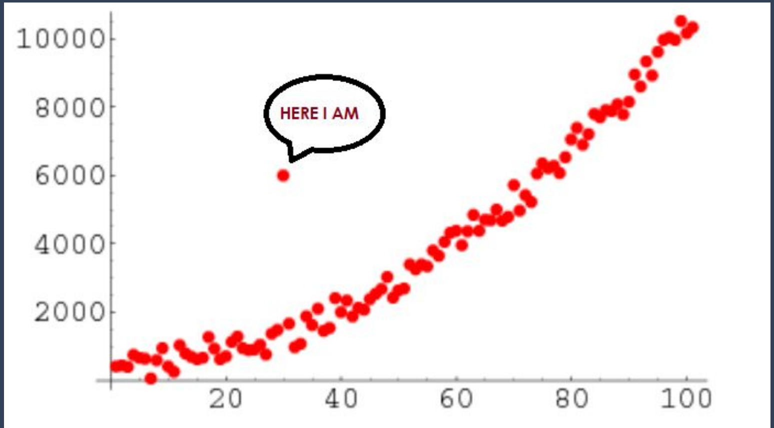
Multivariate Analysis for Titanic Data

MULTIVARIATE ANALYSIS
How did age, ticket class, and gender together influence survival rates? - Analyzes the combined effect of age, ticket class, and gender on survival to identify more complex patterns.
How did ticket price, class, and where passengers boarded collectively impact survival? - Investigates the combined influence of the amount paid for the ticket, class, and boarding location on survival chances.

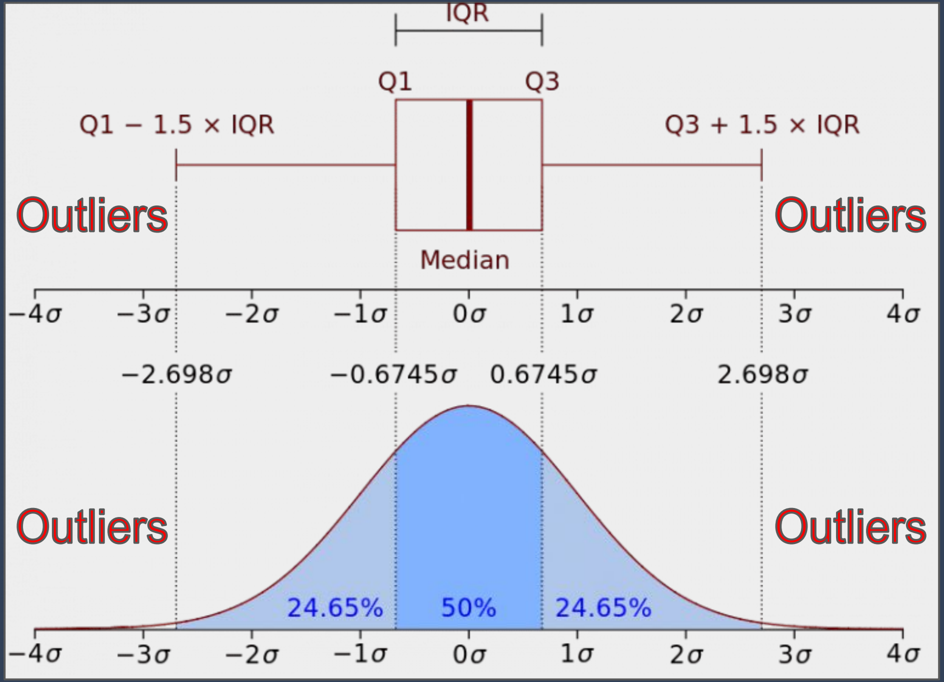


Detecting Outliers

- An outlier is an observation that lies an abnormal distance from other values in a dataset.
- Outliers can be caused by data entry errors, experimental errors, or naturally occurring variations.
- Outliers can skew the results of statistical analyses, leading to misleading conclusions or poor model performance in machine learning.



Detecting Outliers

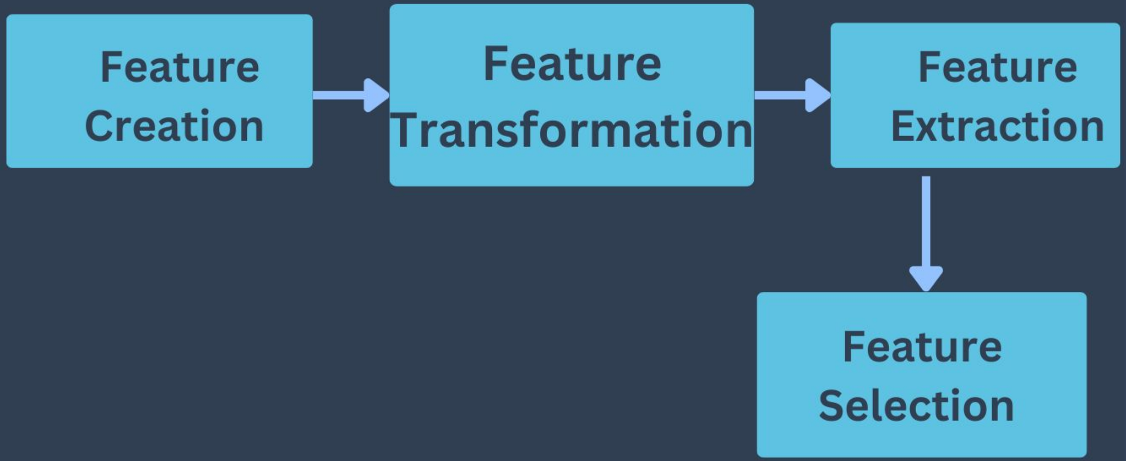


Handling Outliers

- **Remove Outliers:** If outliers are due to errors or irrelevant to the analysis, consider removing them from the dataset.
- **Cap or Floor Values:** Limit outliers by capping extreme values at a set maximum or minimum threshold.
- **Transform Data:** Apply transformations like logarithms or square roots to reduce the impact of outliers.
- **Analyze Separately:** Treat outliers as a separate group if they represent significant but rare events, analyzing them independently from the main dataset.

Feature Engineering

Feature Engineering

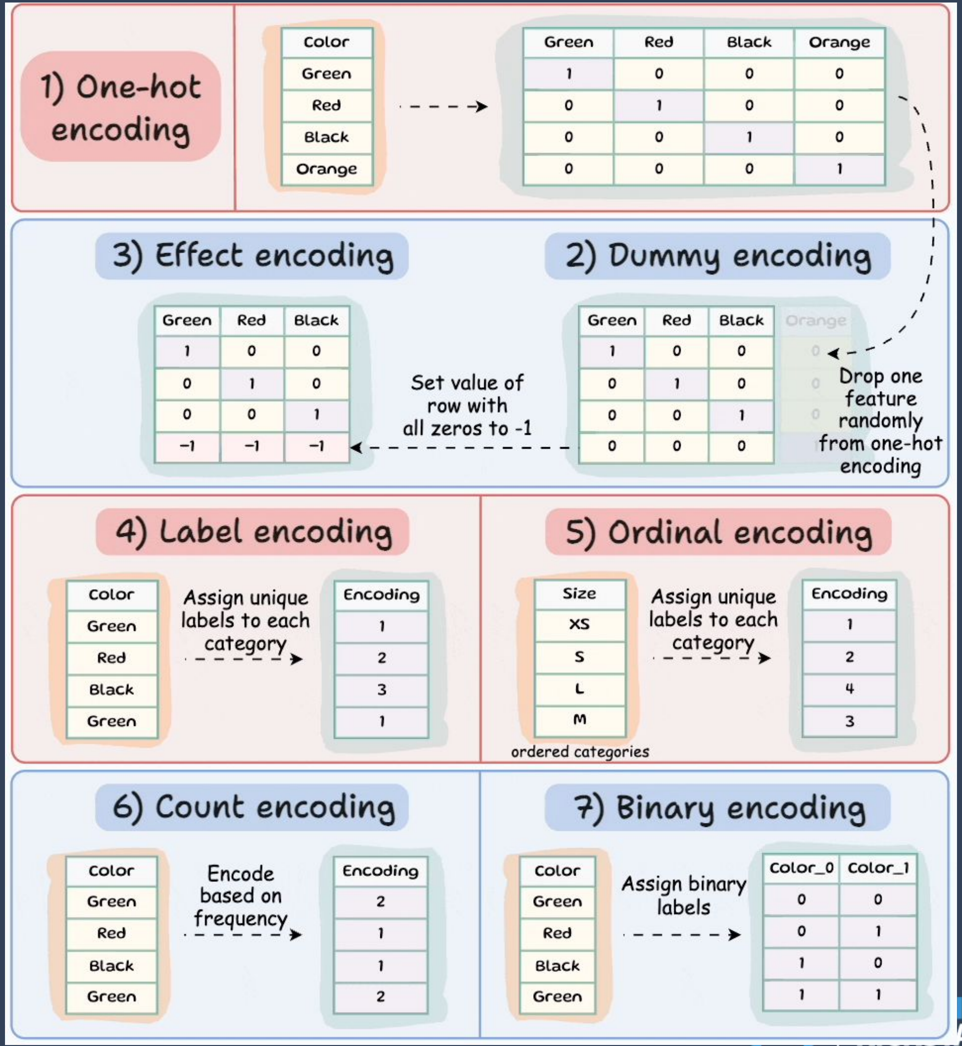


Feature Creation

- Feature creation involves generating new features (variables) from existing data to enhance the predictive power of machine learning models.
- **Techniques:**
  - Combining features (e.g., multiplying or adding two features)
  - Extracting components (e.g., year from a date)
  - Applying domain knowledge to create new features.
- **Examples:** For a dataset with date and time, shall create features like "Day of the Week" or "Hour of the Day" to capture time-related patterns.

Feature Transformation

- Encoding is a feature transformation technique
- Feature encoding is necessary because most machine learning algorithms require numerical input to process data effectively

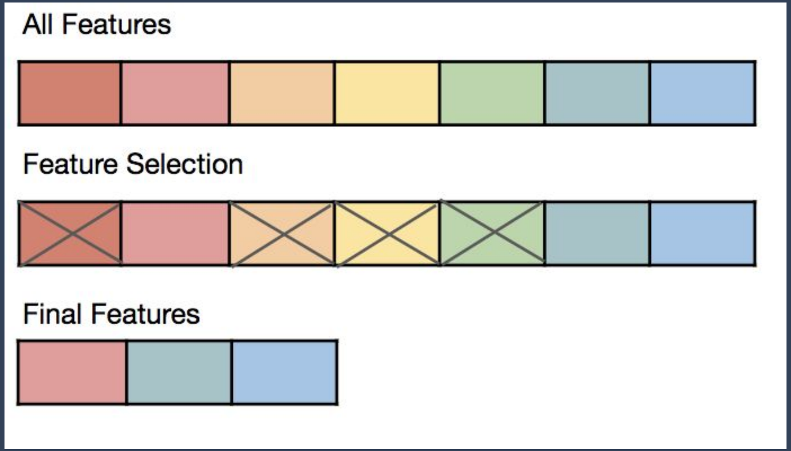


Feature Extraction

- Feature extraction is the process of identifying and selecting the most important information or characteristics from a data set.
- **Techniques:**
  - Dimensionality reduction techniques like Principal Component Analysis (PCA), etc

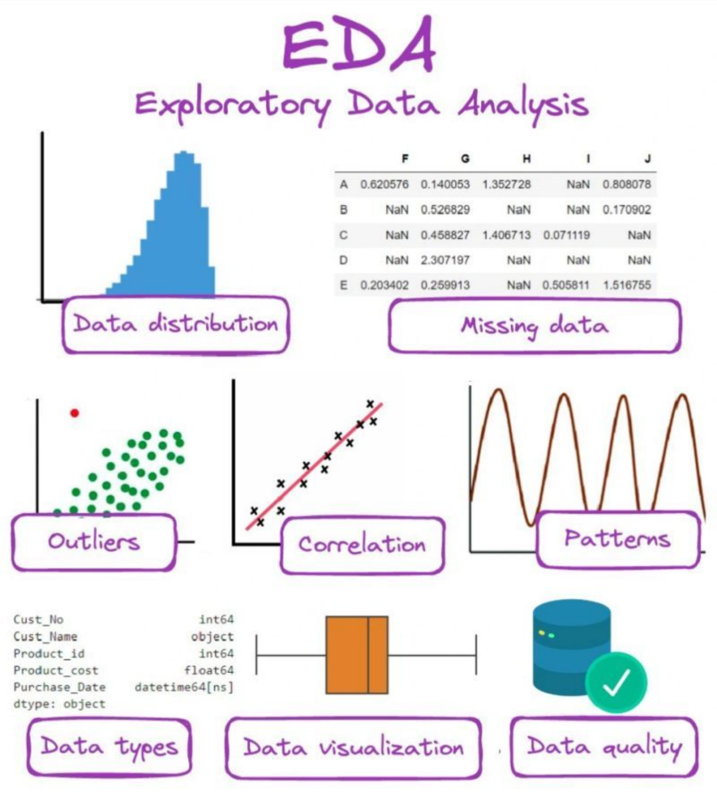
Feature Selection

- Feature selection involves choosing a subset of the most important features from the original dataset to improve model performance by eliminating irrelevant or redundant data.
- **Examples:** In a dataset with 100 features, feature selection might identify that only 20 features are strongly correlated with the target variable and should be used for model training.



Exploratory Data Analysis

- All data processing, visualization & feature engineering techniques discussed so far is collectively referred to as EDA - Exploratory Data Analysis
- Exploratory Data Analysis (EDA) is an umbrella term that encompasses a variety of techniques and processes aimed at understanding and preparing data before modeling.



Summary

01 Data preprocessing includes cleaning, transforming, encoding, and preparing data, which lays the foundation for accurate and effective ML models.

02 Data Cleaning: Essential for ensuring the accuracy and reliability of your dataset by handling missing values, correcting errors, and removing duplicates.

03 EDA: Involves visualizing and summarizing the data to uncover patterns, relationships, and anomalies before modeling.

04 Identifying and addressing outliers is crucial to avoid skewing your analysis and improving model performance.

05 One-Hot Encoding: Transforms categorical variables into binary columns, allowing machine learning models to process them effectively.

Attend the Thursday sessions for more practise!

Get your queries clarified via Wednesday sessions!

Upcoming Class: Intro to Machine Learning

Note: You will get a brief idea of Supervised Machine Learning - Regression , Classification.

Algorithms Covered: Linear Regression, Logistic Regression

Watch the pre-class foundation videos before coming to the next class!

Watch them before the class for better learning outcomes!

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Thank You